Introduction

I built a c++ codebase that simulates 2D collisions and gravitational interactions of n-body problems. The primary use case is for astronomic simulations, hence code needs to handle sufficiently long timeframes and massive objects for gravitational interactions to be meaningful.

Problem statement

The code compiles and runs on my end for simulations of perfectly elastic particles. Gravitational and collision effects work fine for perfectly **elastic** particles, and the conservation of momentum and energy is as expected. When particles however are **inelastic**, the simulation seems to either add/subtract energy from the system, which violates conservation laws.

Work requirement

The work requirement is divided into two separate phases.

Phase 1: Upon acceptance of the offer, the freelancer will have to deliver on the following 4 deliverables within the specified timeframe:

1) Compile and run the code and confirm with me that they are able to produce simulation outcomes for inelastic particles with less than 2% error for the **Testing Sets.**
2) Investigate the problem statement, identify what (combination of) design flaws are causing the violation of conservation laws for elastic particles
3) Provide a written report (2pg max) that explains the key issues found which are likely to result in the observed problem statement. For each issue, a solution should be proposed, with high-level indication where in the code this would need implementing. This will point to specific code parts (eg. force vs potential mismatch, collision/overlap handling) without going into efficiency/design.
4) Provide a written mid-work update when the delivery of phase 1 is roughly halfway through.

Phase 2 (potential but unguaranteed extension): Implementation of the above diagnosis and solution to the code, correcting the key issues giving rise to the problem statement and validating the results numerically (e.g. change in total energy) and visually, by confirming "physical" behaviour in simulation outcomes (e.g. GIF).

Threshold for error

The threshold for error is set to 2% error in total energy between start and end of the simulation, for a reasonable simulation timeline (sufficient to observe effects similar to the GIF images contained in the repo).

Testing Sets

The existing testing sets that are confirmed to produce elastic simulations below the threshold error are provided as Test1, Test2, Test3, Test4. Any further tests to be created should maintain this naming convention.

Delivered materials

Upon acceptance of the offer, the freelancer confirms that they understand the shared **GitHub Branche** sufficiently to commence work on the 4 deliverables.

Handover of materials

Any sharing of materials between the two parties can be coordinated either through GitHub or through a data transfer method (e.g. WeTransfer) of the freelancer's choice. The initiative to request critical materials and share (intermediate) deliverables is fully on the side of the freelancer.

Dependencies

The code is dependent on the open source boost_1_86_0 library for high precision floats. This can be shared following instructions in **handover of materials** upon request.

Payment

A payment of 120 USD will be provided on the completion of all 4 deliverables described in Phase 1, in full, within the specified **timeframe.**

A discretionary bonus payment of up to 50 USD can be provided, subject to a high quality of deliverables.

If disagreement arise regarding whether the deliverables satisfy the SOW requirements, both parties will first attempt to resolve the matter in good faith through written discussion within the Fiverr message thread before escalating the matter to Fiverr's Resolution Center.

Timeframe

The timeframe is 8 days from the acceptance of the offer.

Communication requirements

Both parties agree to use Fiverr as primary communication platform.

It is the sole responsibility of the freelancer to timely make any requests for additional information (both guidance regarding the project) and requests for additional files which is deemed necessary for successful delivery of the 4 deliverables.

It is the responsibility of the client to respond within 24 hours following any inquiries from the freelancer.

Github Branche

https://github.com/Braumeister-Stefan/ParticleSimulator2/tree/Fiverr_branch

Diagnostics

Logging capacity currently present in the code (kinetic, potential, heating, total energy, momentum) can be leveraged for purposes of debugging.

Elastic/Inelastic

Elasticity of particles is defined by a restitution parameter with 0 equal to perfect inelasticity and 1 perfect elasticity.